

HDOC – 25th day

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PROBLEM STATEMENT:

Write a C program that displays the following menu and uses a switch statement to print the “Decreasing patterns” selected by the user. Use nested loops without arrays. Each pattern may be implemented in a separate parameter less function with no return value.

Aim:

To write a C program using a switch statement, functions, and nested loops to display different decreasing patterns according to the user's choice, without using arrays.

Algorithm:

1. Start the program.
2. Declare four parameterless functions with no return value:
pattern1(), pattern2(), pattern3(), and pattern4().
3. Display the menu:
 - 3.1) Decreasing right-angled triangle
 - 3.2) Decreasing natural-number pattern
 - 3.3) Decreasing repeated-number pattern
 - 3.4) Reverse-number triangle
 - 3.5) Exit
4. Accept the user's choice.
5. Use a switch statement.
6. For choices 1–4, call the corresponding function.
7. Inside each function:
 - 7.1) Input the number of rows n.
 - 7.2) Use an outer loop for rows.

7.3) Use an inner loop to print the elements in each row.

8. If the user selects 5, exit the program.

9. If the choice is invalid, display "Invalid choice".

10. Stop.

Test Case Table:

Test case	choice	input	Expected output	Actual output	Result
1	1	4	**** *** ** *	**** *** ** *	Pass
2	2	4	1 2 3 4 1 2 3 1 2 1	1 2 3 4 1 2 3 1 2 1	Pass
3	3	4	4 4 4 4 3 3 3 2 2 1	4 4 4 4 3 3 3 2 2 1	Pass
4	4	4	4 3 2 1 3 2 1 2 1 1	4 3 2 1 3 2 1 2 1 1	Pass
5	5	—	Program ended	Program ended	Pass
6	6	—	Invalid choice	Invalid choice	Pass

C Program:



The image shows a screenshot of a terminal window running a C program. The window title is "kali-linux-2026.2-virtualbox-amd64 [Running] - Oracle VirtualBox". The menu bar includes "File", "Machine", "View", "Input", "Devices", and "Help". The toolbar shows icons for file operations and a tab labeled "1". The file name is "*hdoc25.c" and the path is "~/Desktop/HDOC". The code is as follows:

```
1#include <stdio.h>
2
3void pattern1()
4{
5    int n, i, j;
6
7    printf("Enter number of rows: ");
8    scanf("%d", &n);
9    for(i = n; i >= 1; i--)
10    {
11        for(j = 1; j <= i; j++)
12        {
13            printf("* ");
14        }
15        printf("\n");
16    }
17}
18
19void pattern2()
20{
21    int n, i, j;
22
23    printf("Enter number of rows: ");
24    scanf("%d", &n);
25
26    for(i = n; i >= 1; i--)
27    {
28        for(j = 1; j <= i; j++)
29        {
30            printf("%d ", j);
31        }
32        printf("\n");
33    }
34}
35
36void pattern3()
```

```
35
36 void pattern3()
37 {
38     int n, i, j;
39
40     printf("Enter number of rows: ");
41     scanf("%d", &n);
42
43     for(i = n; i >= 1; i--)
44     {
45         for(j = 1; j <= i; j++)
46         {
47             printf("%d ", i);
48         }
49         printf("\n");
50     }
51 }
52
53 void pattern4()
54 {
55     int n, i, j;
56
57     printf("Enter number of rows: ");
58     scanf("%d", &n);
59
60     for(i = n; i >= 1; i--)
61     {
62         for(j = i; j >= 1; j--)
63         {
64             printf("%d ", j);
65         }
66         printf("\n");
67     }
68 }
69
70 int main()
```

kali-linux-2026.2-virtualbox-amd64 [Running] - Oracle VirtualBox

File Machine View Input Devices Help

1 2 3 4

*hdoc25.c
~/Desktop/HDOC

Terminal Emulator
Use the command line

```
73
74 printf("----- Decreasing Patterns -----\\n");
75 printf("1. Decreasing right-angled triangle\\n");
76 printf("2. Decreasing natural-number pattern\\n");
77 printf("3. Decreasing repeated-number pattern\\n");
78 printf("4. Reverse-number triangle\\n");
79 printf("5. Exit\\n");
80
81 printf("Enter your choice: ");
82 scanf("%d", &choice);
83
84 switch(choice)
85 {
86     case 1:
87         pattern1();
88         break;
89
90     case 2:
91         pattern2();
92         break;
93
94     case 3:
95         pattern3();
96         break;
97
98     case 4:
99         pattern4();
100        break;
101
102     case 5:
103         printf("Program ended.");
104         break;
105
106     default:
107         printf("Invalid choice.");
108 }
109
110 return 0;
111 }
```

C Tab Width: 8

Compile the code using:

gcc hdoc25.c -o hdoc25

```
version '1.0' not found
(kali㉿kali)-[~/Desktop/HDOC]
$ gcc hdoc25.c -o hdoc25
(kali㉿kali)-[~/Desktop/HDOC]
$ ./hdoc25
```

Execute the code using:

./hdoc25

```
(kali㉿kali)-[~/Desktop/HDOC]
$ ./hdoc25
——— Decreasing Patterns ———
1. Decreasing right-angled triangle
2. Decreasing natural-number pattern
3. Decreasing repeated-number pattern
4. Reverse-number triangle
5. Exit
Enter your choice: 1
Enter number of rows: 4
* * * *
* * *
* *
*
*
```

```
(kali㉿kali)-[~/Desktop/HDOC]
$ ./hdoc25
——— Decreasing Patterns ———
1. Decreasing right-angled triangle
2. Decreasing natural-number pattern
3. Decreasing repeated-number pattern
4. Reverse-number triangle
5. Exit
Enter your choice: 2
Enter number of rows: 4
1 2 3 4
1 2 3
1 2
1
```

```
(kali㉿kali)-[~/Desktop/HDOC]
$ ./hdoc25
——— Decreasing Patterns ———
1. Decreasing right-angled triangle
2. Decreasing natural-number pattern
3. Decreasing repeated-number pattern
4. Reverse-number triangle
5. Exit
Enter your choice: 3
Enter number of rows: 4
4 4 4 4
3 3 3
2 2
1
```

```
(kali㉿kali)-[~/Desktop/HDOC]
$ ./hdoc25
——— Decreasing Patterns ———
1. Decreasing right-angled triangle
2. Decreasing natural-number pattern
3. Decreasing repeated-number pattern
4. Reverse-number triangle
5. Exit
Enter your choice: 4
Enter number of rows: 4
4 3 2 1
3 2 1
2 1
1
```

```
(kali@kali)-[~/Desktop/HDOC]
$ ./hdoc25
——— Decreasing Patterns ———
1. Decreasing right-angled triangle
2. Decreasing natural-number pattern
3. Decreasing repeated-number pattern
4. Reverse-number triangle
5. Exit
Enter your choice: 5
Program ended.
```

```
(kali@kali)-[~/Desktop/HDOC]
$ ./hdoc25
——— Decreasing Patterns ———
1. Decreasing right-angled triangle
2. Decreasing natural-number pattern
3. Decreasing repeated-number pattern
4. Reverse-number triangle
5. Exit
Enter your choice: 6
Invalid choice.
```